

Lecture 9

Review for Exam 1

Numbers and codes · Boolean algebra and canonical forms · K-maps and the tabular method · NAND and NOR circuits · timed practice set

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapters 1–3 (review)

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

$$F = A'B + AB' = A \oplus B$$

Today's roadmap

1

§1.3

Numbers, codes

Lectures 1–2

2

§2.4

Algebra and forms

Lectures 3–5

3

§3.3

Minimization

Lectures 6 and 8

4

§3.6

Implementation

Lecture 7

By the end of class you can...

- Convert numbers and do signed arithmetic with overflow checks
- Encode with BCD, excess-3, Gray and parity
- Simplify algebraically and write Σ and Π forms
- Minimize with K-maps (SOP and POS) and the tabular method
- Draw NAND-NAND and NOR-NOR circuits

Agenda for today's session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; Homeworks 4 and 5 collected	—
0:05 – 0:15	What the exam covers; notation; common errors	—
0:15 – 0:35	Conversions, signed arithmetic, codes	Reviews 1–3
0:35 – 0:50	Simplification, complements, canonical forms	Reviews 4, 5
0:50 – 1:00	Break	
1:00 – 1:25	K-map with don't-cares, POS, tabular method	Reviews 6–8
1:25 – 1:35	NAND implementation	Review 9
1:35 – 1:50	Timed practice set and solutions	Practice set

What Exam 1 covers

Lecture	Topic	You should be able to
1	Number systems	convert bases; complements; signed arithmetic; overflow
2	Codes and binary logic	BCD, excess-3, Gray, ASCII, parity; timing diagrams
3	Boolean algebra	prove and simplify; DeMorgan; complements
4	Canonical and standard forms	Σ and Π forms; SOP and POS circuits; the 16 functions
5	Logic families	fan-out, noise margin, power, delay
6	Karnaugh maps	2–4 variables; primes; don't-cares; POS
7	NAND, NOR, XOR	two-level and multilevel conversion; parity
8	Tabular method	prime implicants; chart; Petrick
10	Combinational design	analysis and design procedure

Notation used on the exam

Write it this way

Complement: A' . AND: AB . OR: $A + B$. XOR: $A \oplus B$.

Sum of minterms: $F(A, B, C) = \Sigma m(1, 4, 5)$

Product of maxterms: $F = \Pi M(0, 2, 3, 6, 7)$

Don't-cares listed separately: $+ d(\dots)$

The first variable listed is the MSB of the index.

Tag every base: $(247.375)_{10} = (11110111.011)_2$

Common homework errors

Reading division remainders top-down (it is bottom-up).

Forgetting to drop the end carry in 2's-complement subtraction.

Dropping parentheses when complementing.

K-map columns in binary order instead of Gray order.

Missing the wrap-around groups on a 4-variable map.

Leaving a lone literal un-complemented in NAND-NAND.

Review 1 — Base conversion

§1.3

REVIEW

Convert $(156.625)_{10}$ to binary, octal and hexadecimal.

Group from the binary point

0	1	0	0	1	1	1	0	0	.	1	0	1
2			3			4			.	5		

1	0	0	1	1	1	0	0	.	1	0	1	0
9			C			.	A					

Octal: groups of 3 (top). Hex: groups of 4 (bottom). Pad the ends with 0s.

- 1 Largest power of 2 in 156 is 128: $156 - 128 = 28$
- 2 $28 - 16 = 12$, $12 - 8 = 4$, $4 - 4 = 0 \rightarrow$ 1s at 128, 16, 8, 4: 10011100
- 3 $0.625 \times 2 = 1.25 \rightarrow 1$; $0.25 \times 2 = 0.5 \rightarrow 0$; $0.5 \times 2 = 1.0 \rightarrow 1$
- 4 Join: $(156.625)_{10} = (10011100.101)_2$
- 5 Octal, groups of 3: 010 011 100 . 101 $\rightarrow (234.5)_8$
- 6 Hex, groups of 4: 1001 1100 . 1010 $\rightarrow (9C.A)_{16}$
- 7 Check: $2 \cdot 64 + 3 \cdot 8 + 4 = 156$; $9 \cdot 16 + 12 = 156 \checkmark$

Review 2 — Signed arithmetic

§1.6

REVIEW

In 8-bit 2's complement compute (a) $-75 + 40$ and (b) $-75 + (-60)$. Check for overflow.

(a) and (b) column by column

-75	1	0	1	1	0	1	0	1
+40	0	0	1	0	1	0	0	0
sum	1	1	0	1	1	1	0	1

-75	1	0	1	1	0	1	0	1
-60	1	1	0	0	0	1	0	0
sum	0	1	1	1	1	0	0	1

- 1 75 = 64 + 8 + 2 + 1 = 01001011; invert, add 1: $-75 = 10110101$
- 2 (a) $+40 = 00101000$. Add: $10110101 + 00101000 = 11011101$
- 3 Sign 1: negative. Invert 00100010, add 1: $00100011 = 35$
- 4 So (a) = **-35 ✓** (opposite signs: no overflow)
- 5 (b) $60 = 00111100$; invert, add 1: $-60 = 11000100$
- 6 Add: $1\ 01111001$; drop the end carry: $01111001 = +121$
- 7 Two negatives gave a positive: **overflow**
- 8 True sum $-135 < -128$; $V = C_{in} \oplus C_{out} = 0 \oplus 1 = 1$

Review 3 — Codes and parity

§1.7

REVIEW

(a) 47 in BCD and excess-3. (b) Gray code of 1011. (c) Even-parity bit for 101 1001.

Code	4	7
BCD	0100	0111
Excess-3	0111	1010

1 (a) BCD: $4 = 0100$, $7 = 0111 \rightarrow \mathbf{0100\ 0111}$

2 Excess-3: $4 + 3 = 7 = 0111$, $7 + 3 = 10 = 1010 \rightarrow \mathbf{0111\ 1010}$

3 (b) Copy the leftmost bit: 1

4 XOR neighbours: $1 \oplus 0 = 1$, $0 \oplus 1 = 1$, $1 \oplus 1 = 0 \rightarrow \mathbf{1110}$

5 (c) Count the 1s in 101 1001: four

6 Four is already even: $P = 0$ (odd parity: $P = 1$)

binary	1	0	1	1
Gray	1	1	1	0

Review 4 — Algebra and complements

§2.4

REVIEW

(a) Simplify $xy + xy'z + x'yz$. (b) Find the complement of $F = (A + B'C)D'$.

Tools

$x + x'y = x + y$ (use it in both directions)

DeMorgan from the outside in; keep every parenthesis.

Check (a): majority function, $\Sigma m(3,5,6,7)$.

- 1 (a) $xy + xy'z = x(y + y'z)$ (distributive)
- 2 $= x(y + z) = xy + xz$ ($x + x'y = x + y$)
- 3 $xy + x'yz = y(x + x'z)$ (reuse xy : $x = x + x$)
- 4 $= y(x + z) = xy + yz$ ($x + x'y = x + y$)
- 5 $F = xy + xz + yz$
- 6 (b) $F' = (A + B'C)' + (D')'$ (DeMorgan on the AND)
- 7 $= A'(B'C)' + D$ (DeMorgan, involution)
- 8 $F' = A'(B + C') + D$ (DeMorgan)

Review 5 — Canonical forms

§2.6

REVIEW

Write $F = A'B + AC'$ as a sum of minterms and a product of maxterms.

A B C	A'B	AC'	F
0 0 0	0	0	0
0 0 1	0	0	0
0 1 0	1	0	1
0 1 1	1	0	1
1 0 0	0	1	1
1 0 1	0	0	0
1 1 0	0	1	1
1 1 1	0	0	0

- 1 $A'B$ lacks C : $A'B(C + C') = A'BC' + A'BC$
- 2 $A'BC' = 010 = m_2$ and $A'BC = 011 = m_3$
- 3 AC' lacks B : $AC'(B + B') = AB'C' + ABC' = m_4 + m_6$
- 4 $F = \Sigma m(2,3,4,6)$: the highlighted rows
- 5 Rows where $F = 0$: 0, 1, 5, 7

$$F = \Pi M(0,1,5,7)$$

Break

10 minutes — we resume at the 1:00 mark

Next: K-maps, POS, the tabular method and NAND circuits

Review 6 — K-map with don't-cares

§3.5

REVIEW

Minimize $F(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 10) + d(13)$ as a sum of products.

CD \ AB		CD			
		00	01	11	10
00	m0	1	1		1
	m1		1		
01	m4				
	m5		1		
11	m12				
	m13		X		
10	m8	1			
	m9		1		
	m11				
	m10				1

- 1 Place 1s in 0, 1, 2, 5, 8, 9, 10 and an X in 13.
- 2 Four corners 0, 2, 8, 10: $B = 0, D = 0 \rightarrow B'D'$
- 3 Still uncovered: 1, 5, 9.
- 4 Treat X at 13 as 1: column $CD = 01$ is a quad.
- 5 Quad 1, 5, 9, 13: $C = 0, D = 1 \rightarrow C'D$
- 6 Every 1 is covered: $F = B'D' + C'D$, 4 literals

Review 7 — The same function in POS

§3.4

REVIEW

Write the function of Review 6 as a minimum product of sums.

AB \ CD		CD			
		00	01	11	10
00	m0	1	1	0	1
	m1				
01	m4	0	1	0	0
	m5				
11	m12	0	X	0	0
	m13				
10	m8	1	1	0	1
	m9				

- 1 Write 0s in the empty cells: 3, 4, 6, 7, 11, 12, 14, 15.
- 2 Column $CD = 11$: $C = 1, D = 1 \rightarrow CD$
- 3 Cells 4, 6, 12, 14: $B = 1, D = 0 \rightarrow BD'$
- 4 X at 13 stays a 1: it is not grouped with the 0s.
- 5 So $F' = CD + BD'$
- 6 DeMorgan: $F = (CD)'(BD')' = (C' + D')(B' + D)$
- 7 Check against $\Sigma m(0,1,2,5,8,9,10) \checkmark$

Review 8 — Tabular method

REVIEW

Find the prime implicants and a minimum cover of $F = \Sigma m(0,2,3,7,8,10,11,15)$.

Prime	Term	Covers
-0-0	$B'D'$	0, 2, 8, 10
-01-	$B'C$	2, 3, 10, 11
--11	CD	3, 7, 11, 15

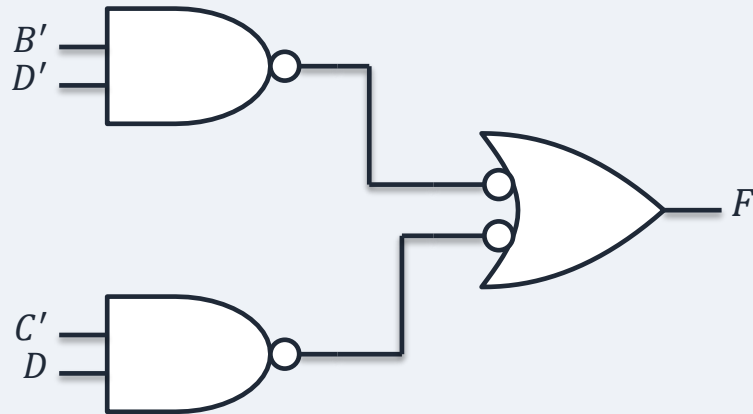
- 1 Group by 1s: 0 | 2, 8 | 3, 10 | 7, 11 | 15
- 2 Pairs: 0,2 0,8 2,3 2,10 8,10 3,7 3,11 10,11 7,15 11,15
- 3 Quads: 0,2,8,10 (-0-0); 2,3,10,11 (-01-); 3,7,11,15 (--11)
- 4 No quads combine: primes $B'D'$, $B'C$, CD
- 5 0 and 8 only in $B'D'$; 7 and 15 only in CD : **essential**
- 6 Together they cover all eight minterms.
- 7 $B'C$ is not needed: $F = B'D' + CD$

Review 9 — NAND implementation

§3.6

REVIEW

Implement $F = B'D' + C'D$ (Review 6) with NAND gates only.



- 1 SOP $B'D' + C'D$: two ANDs and one OR
- 2 Each AND $\rightarrow$ NAND
- 3 The OR $\rightarrow$ invert-OR (a NAND with input bubbles)
- 4 Wires: each output bubble meets an input bubble ✓
- 5 No single-literal term, so no fixes are needed
- 6 Check: $[(B'D')'(C'D)']' = B'D' + C'D$ ✓ (DeMorgan)
- 7 NOR-NOR from Review 7: $[(C' + D')' + (B' + D)']'$

Practice set — 15 minutes

§2.4

YOUR TURN

≈ 15 min

1. Convert $(3AF.8)_{16}$ to decimal.
2. Write -1 and -128 in 8-bit 2's complement.
3. Simplify $(x + y)(x + y') + x'y$.
4. Minimize $F(A, B, C) = \Sigma m(0, 2, 4, 5, 6)$ on a map.
5. How many minterms does the 3-variable odd function have?

Work alone, as on the exam.

Practice set — Solutions 1–3

§2.4

SOLUTION

1. Weights of $3AF$. 8: 256, 16, 1, 1/16
2. $3 \cdot 256 = 768$; $A = 10$: $10 \cdot 16 = 160$; $F = 15$
3. $8/16 = 0.5$. Add: $768 + 160 + 15 + 0.5 = \mathbf{943.5}$
4. $+1 = 00000001$; invert 11111110, add 1: **11111111**
5. $128 = 10000000$; invert 01111111, add 1: **10000000**

6. $3. (x + y)(x + y') = x + yy'$ (distributive)
7. $= x + 0 = x$ (complement, identity)
8. So the expression is $x + x'y$
9. $= x + y$ ($x + x'y = x + y$)

Practice set — Solutions 4–5

§2.4

SOLUTION

BC					
A		00	01	11	10
0	1				1
1	1	1			1

1 4. Place 1s in 0, 2, 4, 5, 6.

2 Outer columns 0, 2, 4, 6 wrap into a quad: only $C = 0 \rightarrow C'$

3 5 is left: pair 4 (100) with 5 (101): $A = 1, B = 0 \rightarrow AB'$

$$F = C' + AB'$$

5 The odd function is 1 when an odd number of inputs is 1:

6 rows 001, 010, 100, 111 $\rightarrow$ 4 minterms, $\Sigma m(1,2,4,7)$

Exam 1: how to prepare

Format

Closed book and notes; calculator and pencil.

Show every step: partial credit follows your work.

Check answers by converting back or substituting a row.

Bring

Pencil, eraser, calculator.

Study plan

Redo every Your turn from Lectures 1–8 without the solution.

Rework the homework problems you missed.

Draw 3- and 4-variable maps until the Gray labels are automatic.

Convert one SOP and one POS circuit to NAND and NOR.

Lecture 10 examples are on the exam too.

Key takeaways

Conversion

divide $\uparrow$, multiply $\downarrow$

Overflow

$$V = C_{in} \oplus C_{out}$$

Canonical

Σ and Π : complementary lists

DeMorgan

$$(x + y)' = x'y'$$

K-map

largest groups, fewest

NAND/NOR

bubbles meet in pairs

Practise until the procedures are automatic.

Homework #6 due at the start of Lecture 13

Next: combinational circuit analysis and design